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Identification and Catalysis of the Potential-Limiting Step in Lithium-Sulfur Batteries

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ABSTRACT

The Li-S chemistry is thermodynamically promising for high-density energy storage but kinetically challenging. Over the past few years, many catalyst materials have been developed to improve the performance of Li-S batteries and their catalytic role has been increasingly accepted. However, the classic catalytic behavior, i.e., reduction of reaction barrier, has not been clearly observed. Crucial mechanistic questions, including what specific step is limiting the reaction rate, whether/how it can be catalyzed, and how the catalysis is sustained after the catalyst surface is covered by solid products, remain unanswered. Herein, we report the first identification of the potential-limiting step of Li-S batteries operating under lean electrolyte conditions and its catalysis that conforms to classic catalysis principles, where the catalyst lowers the kinetic barrier of the potential-limiting step and accelerates the reaction without affecting the product composition. After carefully examining the electrochemistry under lean electrolyte conditions, we update the pathway of the Li-S battery reaction: S_8 solid is first reduced to Li_2S_8 and Li_2S_4 molecular species sequentially; the following reduction of Li_2S_4 to a Li_2S_2 - Li_2S solid with an almost constant ratio of 1:4 is the potential-limiting step; the previously believed Li_2S_2 -to- Li_2S solid-solid conversion does not occur; the recharging reaction is relatively fast. We further demonstrate that supported cobalt phthalocyanine molecules can effectively catalyze the potential-limiting step. After Li_2S_2/Li_2S buries the active sites, it can self-catalyze the reaction and continue driving the discharging process.

INTRODUCTION

The interconversion reaction between S_8 and Li_2S , involving the transfer of 16 electrons per S_8 molecule, promises a large capacity for the cathode of lithium-sulfur (Li-S) batteries. In commonly used ether-based electrolytes, the lithiation (i.e., discharging) of S_8 is believed to go through a series of soluble lithium polysulfides (LPS) to insoluble Li_2S_2 and then to Li_2S solid.¹⁻³ Undesirable migration of LPS causes active material loss and side reactions, limiting the Coulombic efficiency and cycle life of the battery.^{4, 5} One of the earlier strategies to tackle this problem is to employ a sulfur host material that immobilizes LPS.⁶⁻⁹ Later, it was realized that a good sulfur host should also be able to promote the electrochemical conversion of LPS, which led to the emerging of the electrocatalysis concept for Li-S batteries.¹⁰⁻¹³ Thus far, a considerable number of materials, including doped-carbons,^{14, 15} nitrides,^{16, 17} chalcogenides,¹⁸⁻²⁰ phosphides,^{12, 21-23} borides,^{24, 25} and single-atom materials,²⁶⁻²⁸ have been reported as catalysts for Li-S batteries. These materials usually combine LPS affinity with electrical conductivity to enable easier electron transfer and sulfur conversion.

While the catalytic role of these sulfur host materials has been increasingly accepted over the past several years, the classic catalytic behavior, i.e., reduction of reaction barrier, has not been observed for Li-S batteries. Recently, Huang, Sautet, Duan, and coworkers examined the LPS reduction reactions in a rotating disk electrode setting and found the number of transferred electrons can be increased by applying an electrocatalyst, an important step toward unveiling the catalysis.¹⁵ Yet, the effect on the battery remains to be enhancement in capacity, power rate, and cycle life, as in other cases of flooded electrolyte (i.e., high electrolyte/active material ratio) systems.^{10, 11} In our prior work, we demonstrated that the kinetic barriers of the Li-S reaction, i.e., the charging and discharging overpotentials of the battery, are considerably larger under lean electrolyte conditions (i.e., low electrolyte/active material ratio) where the LPS intermediates cannot be sufficiently dissolved.¹³ However, this observation was still based on the overall battery reaction. It remains unknown what specific step is limiting the reaction rate, whether/how it can be catalyzed, and how the catalysis is sustained after the catalyst surface is covered by solid products. Answering these questions is particularly challenging because the reaction of interest here is a complex system that involves multiple steps including possible solid-solution, solution-solution, solution-solid, and solid-solid conversions.^{3, 29}

Herein, we report the first identification of the potential-limiting step, i.e. the step that dictates the

overpotential of an electrochemical reaction, of the Li-S battery reaction and its catalysis under lean electrolyte conditions. We find that the reduction of dissolved Li_2S_4 is the potential-limiting step of the entire battery reaction, which forms a Li_5S_3 solid product (1:4 mixture of Li_2S_2 and Li_2S solids). The previously believed Li_2S_2 -to- Li_2S solid-solid conversion does not occur in the commonly used voltage range. We further show that heterogenized cobalt phthalocyanine (CoPc) molecules can effectively catalyze this step. Compared to a graphitic carbon surface, the strong interaction between the cobalt center and the sulfur of LPS can effectively reduce the energy barrier of Li_2S_4 reduction to solid Li_5S_3 . After the initial deposition, the CoPc active sites are covered by the sulfide solid which can self-catalyze the reduction of Li_2S_4 , continuing to propel the discharging reaction.

RESULTS AND DISCUSSION

We first evaluate the performance of a typical Li-S cell with a carbon nanotube and sulfur composite (CNT-S) cathode and with an electrolyte-to-sulfur (E/S, $\mu\text{L}/\text{mg}$) ratio of 6, a representative value for lean electrolyte conditions under which the LPS intermediates cannot fully dissolve.^{30, 31} Tested at a low current density of $0.2 \text{ mA}/\text{cm}^2$, the cell shows a discharging and charging voltage profile characteristic of the multiple electrochemical transformation reactions described in the classic picture of Li-S batteries (Figure 1a).^{3, 5, 29, 32} During the first-voltage-plateau discharging (2.1-2.4 V vs Li^+/Li), the S_8 molecular solid undergoes a ring-opening reaction to form long-chain Li_2S_8 molecules that are soluble in the ether electrolyte (2.3-2.4 V), which are subsequently reduced to shorter-chain Li_2S_6 and Li_2S_4 molecules sequentially (2.1-2.3 V). X-ray photoelectron spectroscopy (XPS) measurement confirms that the discharging product at the end of the first plateau corresponds to Li_2S_4 (Figure S1). In the second-plateau discharging at 2.1 V, Li_2S_4 is firstly converted to insoluble Li_2S_2 and then reduced to Li_2S solid. The charging, which does not display an apparent voltage plateau, is believed to be the reverse process of the “solid-solution-solid” discharging conversion. At a higher current of $0.8 \text{ mA}/\text{cm}^2$, the cell completely loses the second discharging plateau (Figure 1b), meaning that the deposition of $\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$ does not occur. Note that this current density, corresponding to 0.12 C, is far from the mass transport limiting region for typical batteries and thus serves well for the purpose of kinetic studies. Control experiments suggest that this is caused by a large kinetic barrier associated with the reduction of Li_2S_4 : when we lower the discharging cutoff voltage from 1.7 to 0.8 V, the second discharging plateau can be

recovered with a large initial overpotential of ~ 0.8 V (Figure 1c). We note that this kinetic barrier is only observed under lean electrolyte conditions (Figure S2), which corroborates our earlier conclusion that catalysis, namely reduction of reaction barrier, is most needed for lean-electrolyte Li-S batteries.¹³ The charging process has considerably better kinetics than the discharging, as we find the fully discharged cell can be effectively recharged at high rates up to 4 mA/cm^2 (Figure 1d and S3). Taken together, the potential-limiting step of Li-S conversion is the initial reduction of Li_2S_4 during the discharging.

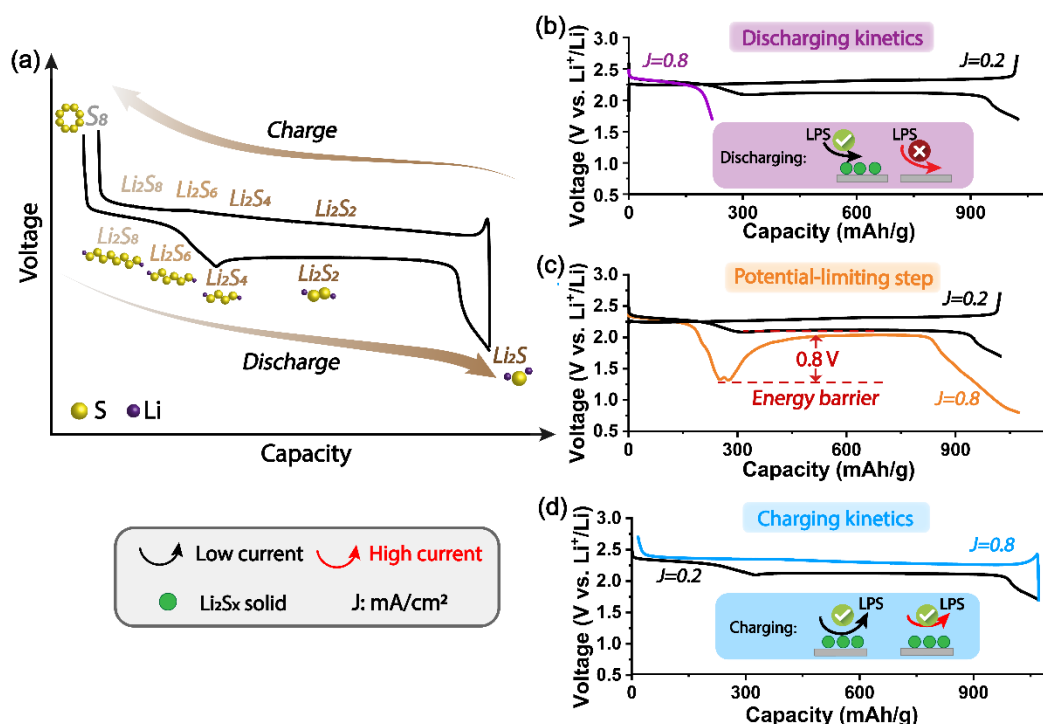


Figure 1. (a) Representative charging-discharging voltage profile of CNT-S electrode at a current density (J) of 0.2 mA/cm^2 under lean electrolyte conditions. Reactants, intermediates and products are labelled at corresponding discharging/charging depths to illustrate the prevalent reaction pathway. Voltage profiles of CNT-S electrode (b) discharged and charged at 0.2 mA/cm^2 (black trace) and discharged at 0.8 mA/cm^2 (purple trace), (c) discharged and charged at 0.2 mA/cm^2 (black trace) and discharged at 0.8 mA/cm^2 to an extended cutoff voltage of $0.8 \text{ V vs Li}^+/\text{Li}$ (orange trace), and (d) discharged at 0.2 mA/cm^2 (black trace) and charged at 0.8 mA/cm^2 (blue trace).

With observation of the potential-limiting step, we seek to catalyze the Li-S battery reaction under

lean electrolyte conditions. We construct an electrocatalyst by supporting CoPc molecules on CNTs (Figure 2a). With molecular-level dispersion of CoPc (Figure S4), this catalyst provides abundant well-defined catalytic sites without considerable alteration of the morphology of the CNT support or the resulting sulfur electrode material (Figure S5). With a low CoPc content of 0.5 wt.% in the CoPc/CNT-S electrode material, the reaction barrier of the potential-limiting step, i.e. the initial reduction of Li_2S_4 , is substantially reduced from 0.80 to 0.02 V at the current density of 0.8 mA/cm^2 (Figure 2b, c). As the current density is increased stepwise to 1.4 mA/cm^2 , the overpotential increases slightly and remains much lower than that without the CoPc catalyst (Figure 2b).

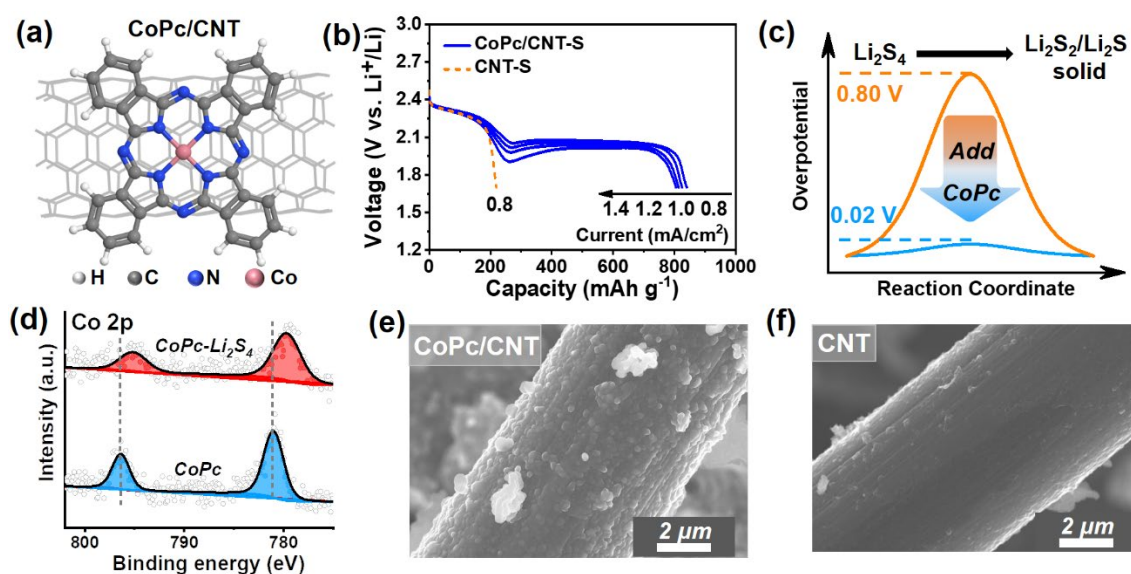


Figure 2. (a) Molecular structure of the CoPc/CNT catalyst. (b) Representative discharging voltage profiles of CNT-S and CoPc/CNT-S electrodes at varied discharging rates under lean electrolyte conditions. (c) Reduction of reaction barrier by CoPc at 0.8 mA/cm^2 . (d) Co 2p XPS spectrum of CoPc before and after interacting with Li_2S_4 . (e, f) SEM images of sulfide nucleation/deposition on CoPc/CNT- and CNT-coated carbon fiber paper.

To characterize the catalytic effect of CoPc molecules, we study the chemical interaction between CoPc and LPS with XPS. We use a Li_2S_4 solution prepared from mixing stoichiometric amounts of S_8 and Li_2S to represent the reactant. After interacting CoPc/CNT and Li_2S_4 , the Co 2p peaks shift to lower binding energies while the S 2p peaks shift to higher binding energies (Figure 2d and S6a), which suggests chemical binding between the Co center of CoPc and the S atoms of LPS. No binding energy

shift is observed in the N 1s spectra (Figure S6b), indicating that the N atoms of the Pc ligand do not bind to the Li of LPS, likely because the latter is solvated by dimethoxyethane. The chemical affinity with LPS allows CoPc-coated CNTs to easily capture Li_2S_4 and reduce it to $\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$. We then assess the ease of depositing sulfide on CoPc/CNT by holding the electrode potential at 2.11 V vs Li^+/Li to reduce Li_2S_4 dissolved in the battery electrolyte.^{33, 34} Compared to CNTs without CoPc, CoPc/CNT shows a considerably higher current response (Figure S7) together with a more uniform and thicker $\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$ deposition layer (Figure 2e, f, Figure S8), indicating easier nucleation and faster growth of solid sulfides enabled by the CoPc catalytic sites. Tracking the solid growth process with scanning electron microscopy (SEM) reveals that the nucleation indeed starts from the CoPc/CNT sites on the carbon fiber paper electrode (Figure S9). Sulfide deposition on a single CoPc-coated CNT is also uniform, as evidenced by subnanometer-resolution mapping of the S element with scanning transmission electron microscopy (Figure S10).

Unlike classic catalysis in the gas or liquid phase where the product leaves the catalyst surface after reaction, the catalysis here in the Li-S battery generates a solid product ($\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$) that covers the catalyst surface. The question, which has not been answered in previous studies, is how the catalysis or discharging continues after all the active sites are buried by $\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$. The neat tubular morphology of our CoPc/CNT hybrid material allows us to easily discern structural changes on the material surface. We first examine electrodes withdrawn from cells that have been discharged for varied capacity into the second voltage plateau (Figure 3a). We observe that the sulfide product initially nucleates on the CoPc/CNT surface (Figure S11) and then continues to grow into a film which buries CoPc/CNT (Figure 3b). At the end of the discharging, the $\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$ film is dense and bulky (Figure 3c). These results suggest that the solid product covers CoPc/CNT at an early stage of the second discharging plateau and the originally active CoPc sites are no longer available for catalyzing the subsequent discharging. We therefore hypothesize that $\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$ solid can form via a self-catalytic process (Figure 3d), i.e., the initially formed sulfide solid can catalyze the subsequent reduction of Li_2S_4 to $\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$.

To test this hypothesis, we switch to the CNT-S electrode to exclude the catalytic effect of CoPc. Having been discharged at a low current of 0.2 mA/cm^2 for a capacity of 200 mAh/g into the second voltage plateau (Figure 3e), the electrode surface is almost fully covered by a layer of sulfide solid with no CNTs exposed (Figure 3f). Interestingly, this electrode can continue to be discharged at a high current

of 0.8 mA/cm² (Figure 3e), a current at which CNT-S cannot be discharged without a pre-deposited sulfide layer (Figure 1b), which unambiguously confirms the self-catalytic function of Li₂S₂/Li₂S. The overall discharging process exhibits negligible overpotential (Figure 3e), and the final discharging product displays a similar dense structure as in the CoPc/CNT-S case (Figure 3g). The same pre-deposition operation can enable the CNT-S electrode to work at an even lower E/S ratio of 4.3 (Figure S12a). Control experiments verify that the pre-deposited sulfide layer promotes the reduction of Li₂S₄ and facilitates the growth of the Li₂S₂/Li₂S solid product itself (Figure S12b-d). We further notice that a partially covered sulfide layer, however, is not sufficient to enable high-current discharging (Figure S13). These results depict a picture of the lithiation of CNT-S at the second discharging voltage plateau (Figure 3h). At the initial stage, Li₂S₂/Li₂S prefers to deposit onto the CNT surface because of its high conductivity. Limited by its high kinetic barrier, this deposition can only proceed at low current density (Figure S14). After the conductive surface is fully passivated by the discharging product, further deposition goes onto the Li₂S₂/Li₂S surface. Discharging can now take place at high current density because the sulfide surface self-catalyzes the reaction, which is likely enabled by the good affinity between Li₂S₄ and Li₂S₂/Li₂S via Li-S interaction. Therefore, CoPc functionalization crucially facilitates strong interaction with Li₂S₄ via Co-S bonding, which matches well with the XPS result (Figure 2d), and thus makes the CNT surface catalytically active for the potential-limiting step, namely Li₂S₄ reduction to Li₂S₂/Li₂S.

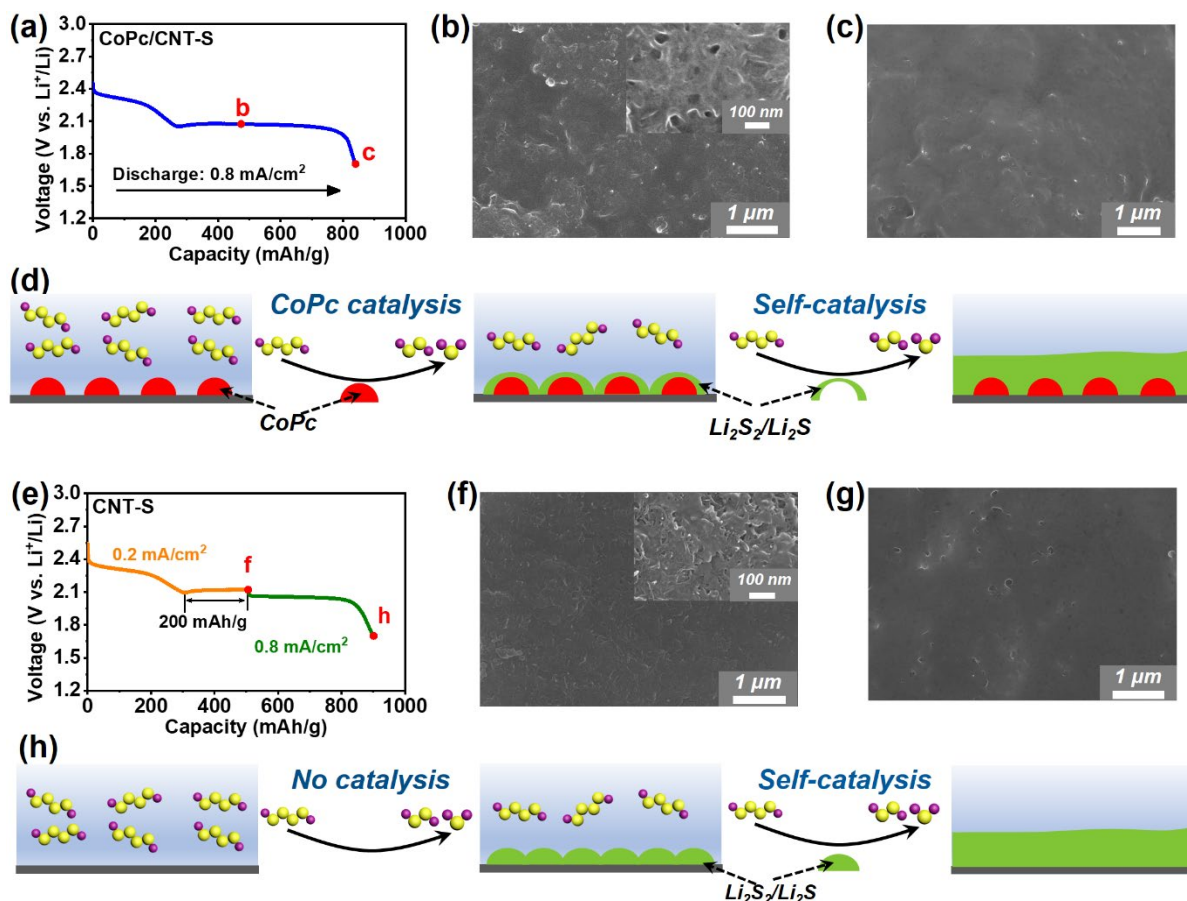


Figure 3. (a) Discharging voltage profile of CoPc/CNT-S electrode at 0.8 mA cm⁻² under lean electrolyte conditions. SEM images of the electrode (b) discharged into the second voltage plateau for 200 mAh g⁻¹ and (c) at the fully discharged state. The inset of (b) is a higher-magnification SEM image. (d) Schematic illustration of Li₂S₂/Li₂S deposition on CoPc/CNT electrode. (e) Discharging voltage profile of CNT-S electrode at 0.2 mA/cm² and then at 0.8 mA/cm² under lean electrolyte conditions. SEM images of the electrode (f) discharged into the second voltage plateau for 200 mAh g⁻¹ and (g) at the fully discharged state. The inset of (f) is a higher-magnification SEM image. (h) Schematic illustration of Li₂S₂/Li₂S deposition on CNT electrode.

The interesting self-catalysis behavior motivates us to investigate the identity of the solid product of the potential-limiting step. The S K-edge X-ray adsorption near edge structure (XANES) spectra of CoPc/CNT-S electrodes fully discharged at 0.2 and 0.8 mA/cm² respectively show that both final discharging products are consisted of Li₂S and Li₂S₂ (Figure 4a). CNT-S fully discharged at 0.2 mA/cm² also has a similar composition, agreeing with the basic catalysis principle that introducing a catalyst

improves the reaction rate but does not change the reaction product. We note that the co-existence of Li_2S and Li_2S_2 in the final discharging product agrees well with recent literature.³⁵ Because of the catalytic effect of CoPc, the fast discharged (0.8 mA/cm^2) CoPc/CNT-S electrode has as uniform coverage of the solid product as the slowly discharged CNT-S electrode (0.2 mA/cm^2), as revealed by X-ray fluorescence imaging (Figure 4b).

We further carry out XPS measurements to quantitatively study the discharge product composition, i.e. the $\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$ ratio. After removing the top surface with sputtering (Figure S15), we find that the solid product only contains Li_2S_2 and Li_2S , and their ratio remains approximately 1:4 at varied discharging depth (Figure 4c). The same $\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$ ratio is observed for CNT-S discharged at 0.2 mA/cm^2 and CoPc/CNT-S discharged at 0.2 or 0.8 mA/cm^2 (Figure S16), suggesting that the electrochemical reaction reaches the same product regardless of the discharging rate and presence of a catalyst. Expectedly, CNT-S discharged at high current, during which process the second-plateau discharging does not occur, contains neither Li_2S_2 nor Li_2S (Figure S17). Based on the final product of 1:4 $\text{Li}_2\text{S}_2:\text{Li}_2\text{S}$, the theoretical capacity of the overall discharging reaction starting from S_8 is calculated to be $\sim 1400 \text{ mAh/g}$. The first discharging plateau, which is S_8 reduction to Li_2S_4 , should account for $\sim 30\%$ of the total capacity. This deduction is in line with the experimental results which show that the two discharging plateaus have a capacity ratio of approximately 3:7 regardless of the discharging rate and presence of a catalyst (Figure S18).

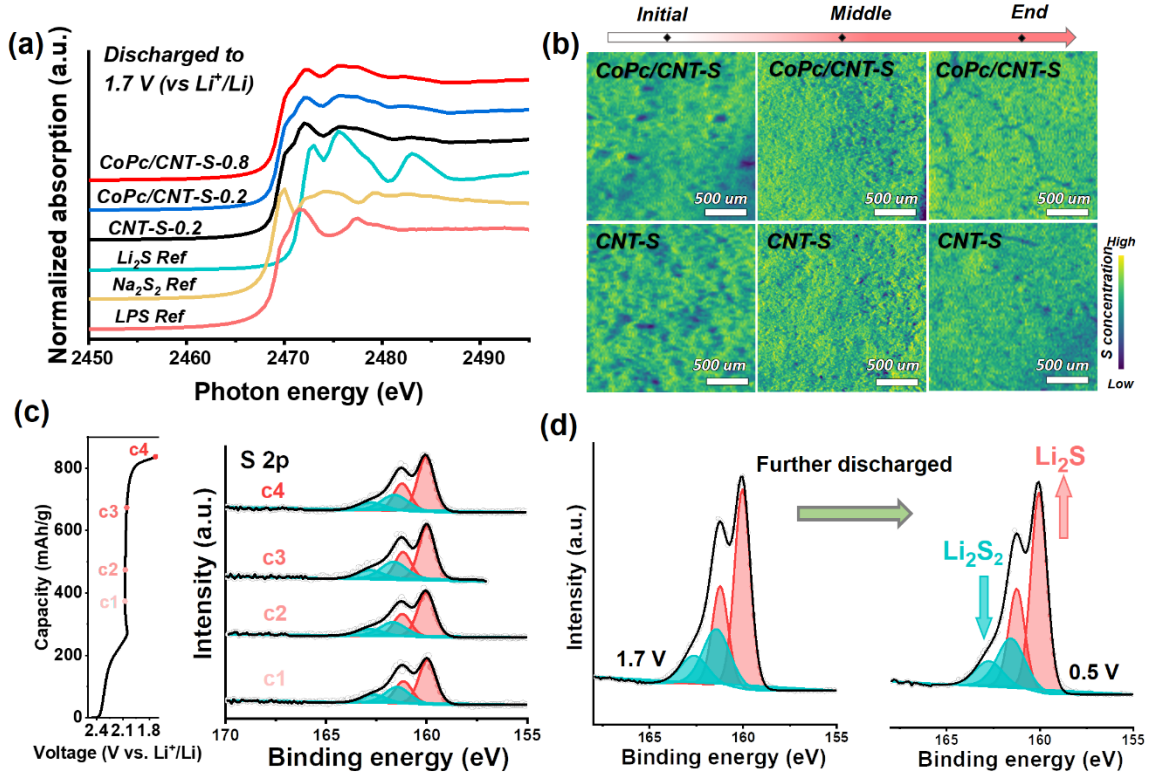


Figure 4. (a) S K-edge XANES spectra of fully discharged CoPc/CNT-S electrode at 0.2 or 0.8 mA/cm² and CNT-S at 0.2 mA/cm². (b) X-ray fluorescence imaging of CoPc/CNT-S and CNT-S electrodes discharged at 0.8 mA/cm² and 0.2 mA/cm² respectively to different depths. (c) S 2p XPS spectra of CoPc/CNT-S electrode discharged at 0.8 mA/cm² to different stages along the second voltage plateau. (d) Change in S 2p XPS spectrum of CNT-S electrode after being further discharged from 1.7 V to 0.5 V vs Li⁺/Li.

Our finding that Li₂S₂ and Li₂S are produced in almost a constant ratio across the second discharging plateau indicates that the conversion of Li₂S₂ to Li₂S in the solid state does not occur, which challenges the prevalent understanding of the Li-S battery chemistry that S₈ is reduced to Li₂S₂ and then to Li₂S.^{1-3, 5, 29, 32} As Li₂S₂ reduction to Li₂S is believed to be well above the 1.7 V discharging cutoff voltage, the absence of this conversion reaction in our batteries suggests that this solid-to-solid transition is likely limited by kinetics. To further validate this analysis, we take a fully discharged CNT-S electrode (cutoff voltage 1.7 V vs Li⁺/Li) and continue to discharge it to 0.5 V. We find that the Li₂S₂/Li₂S ratio of the discharged material decreases from 1:4.2 to 1:4.8 after this additional discharging step (Figure 4d). This verifies that the solid-phase Li₂S₂-to-Li₂S conversion indeed demands a large

overpotential and is difficult to go to completion.

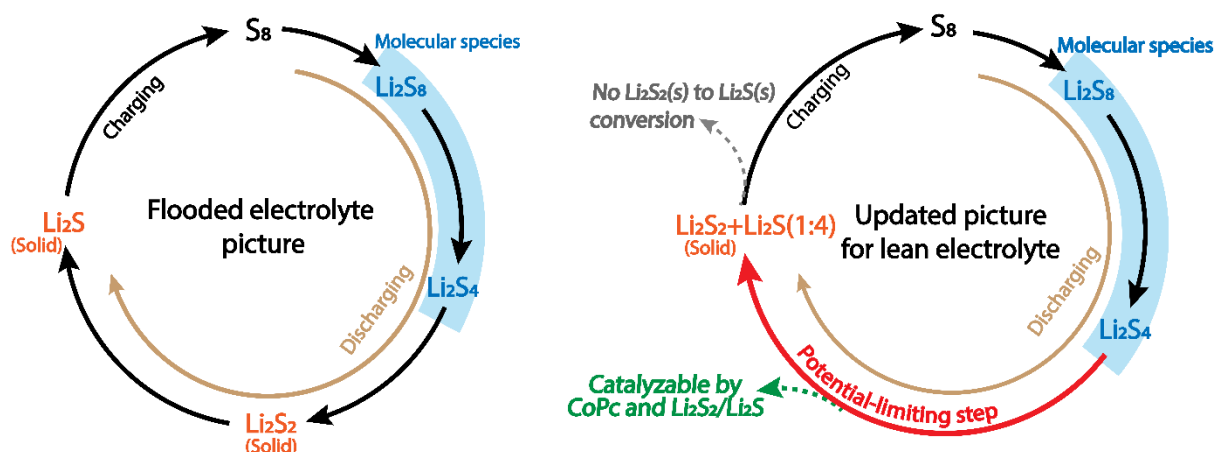


Figure 5. Updated understanding of Li-S battery chemistry under lean electrolyte conditions.

CONCLUSION

The findings in this work allow us to update the understanding of the Li-S battery chemistry under lean electrolyte conditions (Figure 5). In the discharging process, the initial stepwise conversion of S_8 to soluble Li_2S_8 and Li_2S_4 molecular species proceeds with relatively fast kinetics. The following reduction of Li_2S_4 to a solid mixture of Li_2S_2 and Li_2S (1:4) is the potential-limiting step and has a large kinetic barrier on graphitic carbon surfaces. CoPc molecules supported on the carbon surface can effectively catalyze this step and facilitate the initial deposition of Li_2S_2/Li_2S at high current density. As the discharging continues, the CoPc sites are gradually covered with the solid product and lose their catalytic function. However, the discharging does not stop at this point. The Li_2S_2/Li_2S product can self-catalyze this reaction and the product continues to accumulate until the discharging is driven to the end. Compared to discharging, the charging process is much faster and thus not a rate-limiting factor in this system. This work elucidates the catalysis in Li-S batteries and will guide future materials development for accelerating the chemistry under application-relevant conditions.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information including complete experimental details and additional figures is available free of charge.

Experimental details, additional electrochemical data, morphology and composition characterization of materials and electrodes, schematic illustration of sulfide deposition, and calculation of theoretical capacity of Li-S conversion reactions

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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TOC Graphic:

